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IT3081 – Statistical Modelling

Group Assignment Project Guide

Registered topic: Online Retail Transaction Analysis **Proposed consultancy title:** Customer Repurchase and Value Prediction for a UK Online Retailer **Dataset:** Online Retail II – UCI Machine Learning Repository

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1. The Key Decision: Sharpening the Topic and Restructuring the Data

1.1 From a dataset description to a business problem

"Online Retail Transaction Analysis" describes a dataset rather than a business problem. The assignment brief requires an organisational problem with decisions attached, so the project should be framed more sharply as:

Customer Repurchase and Value Prediction for a UK Online Retailer

Client scenario: A UK-based online gift wholesaler wants to know which customers will return, how much those customers are worth, and where it should spend its retention budget.

Confirm the final wording with the lecturer during topic approval.

1.2 The dataset

Online Retail II (UCI Machine Learning Repository) contains approximately one million invoice lines recorded between December 2009 and December 2011. It has eight columns:

Column	Description
Invoice	Invoice number; a leading "C" indicates a cancellation
StockCode	Product code
Description	Product name
Quantity	Units per line (negative for returns/cancellations)
InvoiceDate	Date and time of the transaction
Price	Unit price in GBP
Customer ID	Customer identifier (frequently missing)
Country	Customer's country

Use **Online Retail II** rather than the original *Online Retail* dataset (December 2010 – December 2011 only). The additional year of history is important for Tasks 5 and 9.

1.3 Restructuring to a customer-level table

The raw invoice lines cannot be used directly for ANOVA, regression or PCA. The data must be **aggregated to one row per customer** with engineered features.

To prevent **data leakage**, define a cutoff date (for example, **9 September 2011**):

Features – built only from data before the cutoff:

- Recency, Frequency and Monetary value (RFM)
- Customer tenure
- Average basket value
- Number of distinct products purchased
- Cancellation rate
- Country (UK vs international)
- Whether the customer was first acquired in Q4

Targets – built from the 90 days after the cutoff:

- **Repurchase** (binary): did the customer buy again?
- **Future spend** (continuous): how much did the customer spend?

This design produces a classification problem, a regression problem, genuine comparison groups for inference, and a defensible evaluation strategy. Expect examiners to question leakage during the viva.

2. Task 1 – Understanding the Industry Problem

Framing: Retention economics – retaining an existing customer is typically less costly than acquiring a new one.

Industry background: UK online retail and B2B gift wholesale; many customers are small independent gift shops.

Business decisions affected:

- Which customers should receive retention offers
- How the marketing budget should be allocated
- How Q4 demand and inventory should be planned

Stakeholders:

- Marketing team
- Finance team
- Operations and inventory management
- Wholesale customers (small retailers)

Consultancy objectives (suggested):

1. Describe purchasing behaviour and revenue concentration.
2. Test statistically whether key customer groups differ in value and loyalty.

3. Build and compare models predicting repurchase and future spend.
4. Recommend a practical, profit-aware retention strategy.

3. Task 2 – Research Landscape

3.1 Structure of the review

Organise the literature review around four themes:

1. RFM segmentation
2. Churn and repurchase prediction
3. Customer lifetime value (CLV) modelling
4. Methods: penalised regression, Bayesian approaches, and experimental design

3.2 Anchor papers

Paper	Journal / Venue	Relevance
Chen, Sain & Guo (2012)	<i>Journal of Database Marketing & Customer Strategy Management</i>	Original study using this dataset; supports dataset justification
Fader, Hardie & Lee (2005), "Counting your customers the easy way"	<i>Marketing Science</i>	Probabilistic CLV models (BG/NBD)
Neslin et al. (2006), "Defection detection"	<i>Journal of Marketing Research</i>	Churn prediction
Gupta et al. (2006), "Modeling customer lifetime value"	<i>Journal of Service Research</i>	CLV modelling
Verbeke et al. (2012)	<i>European Journal of Operational Research</i>	Profit-driven churn evaluation
Tibshirani (1996)	<i>Journal of the Royal Statistical Society: Series B</i>	LASSO
Zou & Hastie (2005)	<i>Journal of the Royal Statistical Society: Series B</i>	Elastic net

Paper	Journal / Venue	Relevance
Jolliffe & Cadima (2016)	<i>Philosophical Transactions of the Royal Society A</i>	PCA review
Kohavi et al. (2009)	<i>Data Mining and Knowledge Discovery</i>	Online controlled experiments

3.3 Meeting the minimum requirements

- Minimum **15 peer-reviewed papers**
- Minimum **10 journal papers**
- At least **5 papers from the last five years (2021 onward)**

Suggested Google Scholar searches for recent papers:

- "customer churn e-commerce"
- "customer lifetime value prediction"
- "RFM machine learning"

Important: Verify every reference personally (authors, year, journal, DOI). The academic integrity section of the brief treats fabricated citations as serious misconduct.

3.4 Making the review critical

Build a comparison table with these columns for every paper:

Author (Year)	Method	Dataset	Evaluation Metric	Strengths	Weaknesses / Gaps

3.5 Research gap to claim

Many studies optimise predictive accuracy while neglecting **interpretability**, **calibrated probabilities**, and **profit-based decision-making**. This project addresses all three.

4. Task 3 – Dataset Understanding and Descriptive Analysis

4.1 Data quality issues to document

Issue	Recommended Treatment
Missing Customer IDs (roughly a quarter of rows)	Keep for sales totals; exclude from customer-level models. Justify the choice.
Cancellations (Invoice starts with "C") and negative quantities	Separate out; use to compute cancellation rate
Zero or negative prices	Remove from revenue calculations
Non-product StockCodes (POST, M, D, BANK CHARGES, AMAZONFEE)	Filter out of product analysis
Duplicate rows	Identify and remove
Extreme outliers (very large single orders)	Investigate; consider log transformation or capping

4.2 Descriptive analyses and visualisations

- Summary statistics for Quantity, Price, line revenue and customer-level features
- Histograms and box plots (raw and log scale)
- Revenue by country (highlighting UK concentration)
- **Pareto chart** of cumulative revenue by customer
- Monthly and weekly revenue trends (strong November peak)
- Revenue by day of week and hour of day

4.3 Turning observations into business insights

Every observation should lead to an organisational implication. For example:

"The top X% of customers generate Y% of revenue, so the retailer's income is highly concentrated and exposed to the loss of a small number of buyers."

5. Task 4 – Statistical Inference

5.1 Planned tests

Test Type	Business Question	Method
Comparison of means	Do international customers have a higher average order value than UK customers?	Welch t-test on log order value; Mann–Whitney U as a robustness check
Comparison of proportions	Does the repurchase rate differ for Q4-acquired customers?	Two-proportion z-test or chi-square test
Comparison of variances	Is order value more variable for international customers?	Levene or Brown–Forsythe test (the F-test is too sensitive to non-normality)
ANOVA	Does basket value differ by day of week or by RFM segment?	Welch ANOVA with Games–Howell post-hoc; Kruskal–Wallis as a check

5.2 Required structure for every test

1. State the null and alternative hypotheses.
2. Justify the choice of test (including assumption checks).
3. Present the results (test statistic, degrees of freedom, p-value).
4. Report an **effect size** (Cohen's d, Cramér's V, η^2 , etc.).
5. Interpret the findings.
6. Explain the practical implications for management.

5.3 Key points to emphasise

- Monetary variables are heavily right-skewed. Use log transformations or pair parametric tests with non-parametric alternatives, and explain why.
- With thousands of customers, nearly every difference will be statistically significant. **Statistical significance is not the same as practical significance**, so effect sizes are essential.

6. Task 5 – Predictive Statistical Modelling

6.1 Modelling progression

Repurchase (binary target):

1. **Logistic regression** – interpretable baseline; present odds ratios.

2. **Forward/backward stepwise selection vs LASSO vs elastic net logistic regression** — discuss correlated RFM features; elastic net handles correlated predictors better.

Future spend (continuous target):

3. **Multiple linear regression** on log spend vs **Gamma GLM with a log link**. The Gamma GLM respects positive, skewed outcomes (direct link to LO1).

6.2 Evaluation beyond accuracy

Metric	Purpose
AUC	Ranking ability
Calibration curve and Brier score	Whether predicted probabilities can be trusted
Top-decile lift	How much better targeting the top 10% is than random selection
Expected-profit comparison	Business value of alternative targeting strategies
RMSE / MAE (spend models)	Prediction error on the spend scale

6.3 Assumption checks

- Multicollinearity (Variance Inflation Factor)
- Linearity of the logit
- Residual diagnostics for regression and GLM models
- Influential observations

6.4 Expected recommendation

A **penalised (LASSO or elastic net) logistic regression model**, selected for its interpretability, stability and well-calibrated probabilities rather than raw AUC alone.

7. Task 6 – Critical Evaluation of Experimental Design

7.1 Completely Randomized Design (CRD)

Randomly assign at-risk customers to one of three treatments:

- Discount email
- Reminder email (no discount)
- No contact (control)

Outcome: 90-day repurchase.

7.2 Randomized Complete Block Design (RCBD)

Block by **RFM segment** or **UK vs international**, then randomise treatments within each block. This removes variation caused by very different customer types and produces more precise treatment effect estimates.

7.3 Limitations and implementation challenges

- **Seasonality:** Q4 effects confound results depending on when the test runs.
- **Contamination:** Wholesale customers may communicate with each other.
- **Opportunity cost:** Withholding offers from the control group has a business cost.
- **Sparse data:** Many customers place very few orders.
- **Ethics:** Treating customers unequally raises fairness questions.

Supporting literature: Kohavi et al. (2009) on online controlled experiments.

8. Task 7 – Critical Evaluation of Principal Component Analysis

Take a balanced, two-sided position.

8.1 Where PCA is not worthwhile

For the customer-level model:

- Only around 10 features — dimensionality reduction is not necessary.
- Management needs to interpret coefficients such as "recency"; principal components destroy that interpretability.
- LASSO and elastic net already address correlated predictors.

8.2 Where PCA is useful

For the **customer × product purchase matrix** (thousands of StockCodes):

- Reveals latent buying patterns (e.g. "Christmas decor buyers" vs "kitchenware buyers").
- Supports segmentation and product recommendation.

- Reduces sparsity and noise.

8.3 Expected impact on predictive modelling

Likely little or no improvement for the customer-level model, with a loss of interpretability. Component scores from the product matrix could, however, be tested as additional predictors.

Supporting literature: Jolliffe & Cadima (2016).

9. Task 8 – Critical Evaluation of Bayesian Statistical Methods

9.1 Naïve Bayes

- **Advantages:** Fast, simple, works with small samples.
- **Limitation:** The independence assumption is clearly violated because RFM variables are correlated.
- **Discussion point:** Probabilities may be poorly calibrated, yet customer rankings can still be reasonable.

9.2 Bayesian (hierarchical) regression

- Many countries have only a handful of customers.
- **Partial pooling** produces stable country-level estimates where classical estimates are unreliable.
- This is the strongest case where Bayesian methods can outperform traditional approaches.

9.3 Bayesian decision making

Send a retention offer only when:

$$P(\text{repurchase} \mid \text{offer}) \times \text{margin} > \text{cost of offer}$$

Mention the **BG/NBD model** (Fader, Hardie & Lee, 2005) as the industry-standard probabilistic approach to CLV.

9.4 Limitations

- Choice of priors can be contested.

- Higher computational cost (MCMC).
- Harder to explain to non-technical stakeholders.

10. Task 9 – Time Series Analysis

Although not required to fit models, this dataset contains a genuine time dimension. A simple decomposition plot of weekly revenue will strengthen the discussion.

10.1 Discussion points

Component	Observation / Application
Trend	Long-term growth or decline in weekly revenue
Weekly seasonality	No Saturday orders – explain this operational pattern
Annual seasonality	Strong Q4 / November peak
Forecasting	Weekly demand forecasts
ARIMA / SARIMA	Model trend and seasonal structure

10.2 Honest limitation

Two years of data is thin for reliably estimating annual seasonality (only two seasonal cycles).

10.3 Business applications and benefits

- Inventory planning before Q4
- Warehouse and staffing decisions
- Campaign timing
- Cash-flow planning

11. Task 10 – Industry Innovation Proposal

11.1 Proposed solution: Customer Retention Early-Warning System

11.2 Conceptual framework



This framework connects Tasks 5, 6 and 8, giving the whole project a coherent narrative.

11.3 Proposal elements to cover

- **Business need:** Revenue concentration and unmanaged customer attrition
- **Expected benefits:** Targeted spending, higher retention, measurable campaign ROI
- **Implementation challenges:** Data quality, integration, staff adoption, model drift

- **Required resources:** Data pipeline, analytics staff, dashboard tool, CRM/email integration
 - **Organisational impact:** Shift from intuition-based to evidence-based retention decisions
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12. Task 11 – Industry Expert Validation

12.1 Who to contact

Someone in e-commerce or digital marketing in Sri Lanka, for example:

- A manager at a local online retailer
- A CRM or marketing analyst
- A SLIIT industry contact

12.2 Suggested interview questions

1. Is this problem realistic for an online retailer?
2. Is a 90-day repurchase window sensible?
3. What does a typical retention offer cost?
4. What would block implementation of this system?
5. How would management act on the model's outputs?

12.3 Evidence to collect

Interview summaries, meeting minutes, emails, screenshots of online meetings, or completed feedback forms.

12.4 Showing influence on recommendations

Document how feedback changed the recommendations. For example, if the expert says discounts erode margins, shift the recommendation toward non-discount reminders.

13. Task 12 – Final Consultancy Recommendations

Organise recommendations into five groups:

Category	Examples
Strategic	Prioritise retention of high-value customers; adopt profit-based targeting
Operational	Deploy the early-warning system; plan inventory ahead of Q4
Risk management	Monitor model drift; reduce dependence on top customers
Ethical	GDPR compliance (UK-based client); fairness in differential treatment; transparency in automated targeting
Future research	Controlled experiments; product-level PCA segmentation; time series forecasting

Every recommendation must be traced to:

- A specific statistical result
- A research paper
- Industry expert feedback

14. Team Structure and Timeline

14.1 Role allocation (4 members)

Member	Responsibilities	Tasks
A	Data cleaning, feature engineering, descriptive analysis, time series	3, 9
B	Statistical inference, experimental design	4, 6
C	Predictive modelling, PCA, Bayesian methods	5, 7, 8
D	Problem framing, literature management, expert interview, innovation proposal	1, 2, 10, 11
All	Final recommendations, presentation, viva preparation	12

14.2 Timeline

Week	Milestone
1	Topic approval; data cleaning
2	Descriptive analysis; statistical inference
3	Predictive modelling
4	Expert interview; critical evaluation sections (Tasks 6–9)
5	Slides; presentation and viva rehearsal

15. Viva Preparation Checklist

The viva carries **60%** of the marks. Every member should be able to explain:

- ☐ Why the topic was reframed as a repurchase/value prediction problem
- ☐ The cutoff-date design and how it prevents data leakage
- ☐ How missing Customer IDs and cancellations were handled
- ☐ Why Welch tests and non-parametric checks were used
- ☐ The difference between statistical and practical significance
- ☐ Why a Gamma GLM suits the spend outcome
- ☐ Why the penalised logistic model was recommended over alternatives
- ☐ What calibration, lift and expected profit show that AUC does not
- ☐ When PCA should and should not be used for this data
- ☐ Where Bayesian methods could outperform classical methods
- ☐ How the early-warning system connects the modelling, experimental and Bayesian components
- ☐ How expert feedback changed the final recommendations

Presentation format: 15-minute consultancy presentation to the Board of Directors, followed by a 10-minute viva. Every group member must participate in both.